SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA51
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I Mohammad Ilyas (192325017) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n $\phi(n)$\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i. Calculate total encryption time for each algorithm
ii. Analyze trade-offs between security and performance
iii. Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison
Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

		strong justification			
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

2. Block cipher Mode evaluation

Given,

Plain text blocks : P = [1010, 1010, 1111, 1010]

Key : K = 1100

Initialization vector (IV) = 0000

a) Encrypt using ECB and CBC Models

$E_K(X) = X \oplus K$

i) ECB

formula: $C_i = E_K(P_i) = P_i \oplus k$.

Calculations

Plain text	Key	XOR operation	cipher text
1010	1100	1010 ⊕ 1100	0110
1010	1100	1010 ⊕ 1100	0110
1111	1100	1111 ⊕ 1100	0011
1010	1100	1010 ⊕ 1100	0110

∴ C = [0110, 0110, 0011, 0110]

ii) CBC

formula : $C_i = E_K(P_i \oplus C_{i-1})$

Where $C_0 = IV = 0000$

Block 1 : $P_1 \oplus IV$: 1010 ⊕ 0000 = 1010

C_1 = 1010 ⊕ 1100 = 0110

Block 2 : $P_2 \oplus C_1$: 1010 ⊕ 0110 = 1100

C_2 = 1100 ⊕ 1100 = 0000

Block 3 : $P_3 \oplus C_2 = 1111 \oplus 0000 = 1111$

$C_3 = 1111 \oplus 1100 = 0011$

Block 4 : $P_4 \oplus C_3 = 1010 \oplus 0011 = 1001$

$C_4 = 1001 \oplus 1100 = 0101$

$\therefore C = [0110, 0000, 0011, 0101]$

b) Comparision of cipher text outputs.

Feature	ECB	CBC
Encryption	Encrypts each block independently	Uses previous ciphertext block
IV requirement	NA	required
Output	0110, 0110, 0011, 0110	0110, 0000, 0011, 0101
Repeated plaintext	produces repeated ciphertext	Can produce different ciphertext block
Pattern hiding	Poorer	Better.

c) Which mode hides patterns better?

CBC mode hides patterns better than ECB mode.

In ECB mode, identical plaintext blocks produce identical ciphertext blocks because each block is encrypted independently.

In CBC mode, each block depends on the previous ciphertext block, so identical plaintext blocks can produce different ciphertext blocks.

∴ CBC provides better than ECB, provided that an appropriate IV is used. In real applications, a secure, authenticated encryption mode is generally preferred.

4) Diffe-Hellman Key Exchange Analysis

Given,

Prime number $P = 23$

Generator $g = 5$

User A's private key $a = 6$

User B's Private Key $b = 15$

a) Compute public keys of A and B

$$A = g^a \bmod p$$

$$B = g^b \bmod p$$

Here A and B represents the public keys

Public Key of user A

$g = 5,\ a = 6\ p = 23$

$$A = 5^6 \bmod 23$$

$$5^2 = 25 = 2 \pmod{23}$$

$$5^4 = 2^2 = 4 \pmod{23}$$

$$5^6 = 5^4 \times 5^2$$

$$A = (4 \times 2) \bmod 23$$

$$A = 8$$

∴ User A's public Key is 8

Public Key of User B

Given, $g = 5, b = 15; P = 23$

$$B = 5^{15} \bmod 23$$

Using modular exponentiation:

$$5^5 \bmod 23 = 20$$

$$5^{10} \bmod 23 = 9$$

system to distrust

$\therefore 5^{15} \bmod 23 = (5^{10} \times 5^5) \bmod 23$

$= (9 \times 20) \bmod 23$

$= 180 \bmod 23$

$B = 19$

$\therefore$ User B's public key is 19

b) Compute shared secret key

Both users independently calculate the shared secret using others users public key

$K_A = B^a \bmod P$

$K_B = A^b \bmod P$

Shared key calculated by User A

$K_A = 19^6 \bmod 23$

$19^2 = 361 \equiv 16 \pmod{23}$

$19^4 = 16^2 = 256 = 3 \pmod{23}$

$19^6 = 19^4 \times 19^2$

$K_A = (3 \times 16) \bmod 23$

$K_A = 48 \bmod 23 = 2$

Shared key calculated by User B

$K_B = 8^{15} \bmod 23$

$\therefore 8^2 \bmod 23 = 18, 8^4 \bmod 23 = 2$, and $8^8 \bmod 23 = 4$

$8^{15} = 8^8 \times 8^4 \times 8^2 \times 8$

$K_B = (4 \times 2 \times 18 \times 8) \bmod 23$

$K_B = 1152 \bmod 23 = 2$

Thus, $K_A = K_B = 2$

Both users obtain the same shared secret key, which can be used as basis for deriving a symmetric encryption key.

c) Demonstrate the Man in the middle attack

It occurs when an attacker secretly intercepts and modifies communications btw two users.

User A
sends public key A=8
↓
Attacker mallory intercepts the key
↓
mallory (Attacker)
Replaces the public key and establishes seperate secretly with A and B.
↓
User B
Receives the attacker's substituted public key instead of A's genuine key.

Steps involved:

1. User A sends public key A=8 to user B
2. Mallory intercepts A's public key and sends different public key to B
3. User B sends public key B=19 to user A, but Mallory intercepts it and sends as substituted key to A
4. Mallory establishes one shared secret with A and another B
5. Mallory can decrypt messages received from one user, modify them if desired and encrypt them again for the other user.

d) Suggest a secure method to prevent the attack

1. Digital certificates: Use certificates issued by trusted certificate Authority to verify the identities of communicating parties.

2. Digital Signatures : Sign the Diffe-Hellman public key exchange to detect unauthorized modifications.

3) Authenticated TLS : Use TLS with certificate validation to authenticate the server and protect the key exchange

4) Public key verification: Verify the other party's public key through trusted channel.

Conclusion:

Diffie Hellman securely establishes a shared secret when used correctly, but basic Diffe-Hellman alone does not authenticate the users. Authenticated key exchange, such as properly configured TLS, helps prevent Man in Middle attacks.